# IBME Quiz Bank - All 80 Questions with Answers

Innovation, Business Models and Entrepreneurship (NPTEL noc26_mg145), Weeks 1-8. Compiled from the official NPTEL weekly solution PDFs and verified against portal accepted answers. This is a study compilation, not an official NPTEL document.

## Week 1

1. _____is the specific tool of entrepreneurs, the means by which they exploit change as an opportunity for a different business or a different service."

A. Design
B. Innovation
C. Marketing
D. Communication

**Answer: B**

Drucker's line: innovation is the specific tool of entrepreneurs. They use it to turn change into a new business or service.

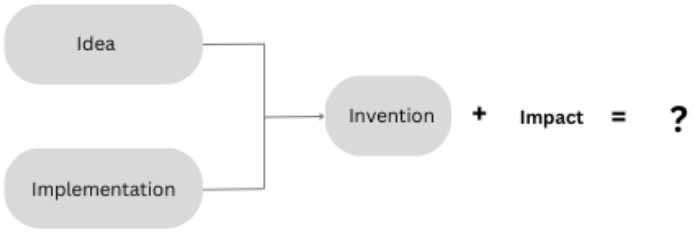
2. Which of the following are the three important I's in the innovation process? (select all that apply)

A. Idea
B. Integration
C. Implementation
D. Industry
E. Impact

**Answers: A, C, E**

The three I's are Idea, Implementation and Impact. An idea that is implemented is an invention. An invention that creates impact is an innovation.

3. As discussed in class, what would be the answer to this equation?




A. Design
B. Innovation
C. Marketing
D. Management

**Answer: B**

Idea + Implementation = Invention. Invention + Impact = Innovation. So the answer is Innovation.

4. Select the correct statement/s.
Statement 1: Innovation begins with curiosity.
Statement 2: Innovation is always a smooth linear process.

A. Both statements are false.
B. Statement 1 is True, and Statement 2 is false.
C. Both statements are true.
D. Statement 1 is False, and Statement 2 is True.

**Answer: B**

Statement 1 is true: innovation begins with curiosity (asking why and what if). Statement 2 is false: innovation is messy and iterative, with loops, failures and restarts, not a smooth line.

5. How do entrepreneurs represent resistance to change? (select all that apply)

A. Ignoring the Idea

B. Ignoring the chances of failure
C. Inertia
D. Ability to exude confidence.

**Answers: A, C**

Resistance to change shows up as ignoring the idea (not even looking at it) and as inertia (sticking to the usual way).

6. Select the correct statement/s.
Statement 1: Frugal innovation sees resource constraints not as a liability but as an opportunity.
Statement 2: Frugal organizations don't seek to wow customers with technically sophisticated products but instead strive to create good-quality solutions that deliver the greatest value to customers at the lowest cost.

A. Both statements are false.
B. Statement 1 is True, and Statement 2 is false.
C. Both statements are true.
D. Statement 1 is False, and Statement 2 is True.

**Answer: C**

Both are true. Frugal innovation treats resource limits as an opportunity. Frugal firms aim for good quality at the lowest cost, not for fancy technology.

7. Which among the following can be considered as the breakthrough innovations? (select all that apply)

A. Development of steam engine
B. Development of wheel
C. Invention of fire
D. Evolution of mobile technology from 2G to 3G

**Answers: A, B, C**

Breakthrough (radical) innovations are new-to-the-world shifts: the steam engine, the wheel and the invention of fire.

8. Select the INCORRECT statement.

A. Design thinking is solution focused.
B. Design thinking is action-oriented.
C. Design thinking is human-centered.
D. Design thinking is product-centered.

**Answer: D**

The question asks for the INCORRECT statement. Design thinking is human-centered, not product-centered. It starts from people's needs.

9. Match the thinking hats in Table 1 with the perspective these represent in Table 2.

| | |
|---|---|
| I | White |
| II | Black |
| III | Red |
| IV | Yellow |
| V | Green |
| VI | Blue |

Table 1

| | |
|---|---|
| A | Feelings, Intuition |
| B | Manage Process |
| C | Information |
| D | Barriers |
| E | Creativity, New Ideas |
| F | Values and benefits |

Table 2

A. I – C, II – D, III – A, IV – F, V – E, VI – B
B. I – D, II – B, III – F, IV – A, V – C, VI – E
C. I – C, II – A, III – F, IV – D, V – B, VI – E
D. I – D, II – F, III – A, IV – E, V – C, VI – B

**Answer: A**

Six Thinking Hats: White = Information, Black = Barriers (caution), Red = Feelings and intuition, Yellow = Values and benefits, Green = Creativity and new ideas, Blue = Manage the process. That is I-C, II-D, III-A, IV-F, V-E, VI-B.

10. "Innovators face unknown unknowns" is the best description of which of the VUCA element?

A. Complexity
B. Volatility
C. Ambiguity
D. Uncertainty

**Answer: C**

Ambiguity means you do not know the cause and effect or even what you do not know. That is 'unknown unknowns'.

## Week 2

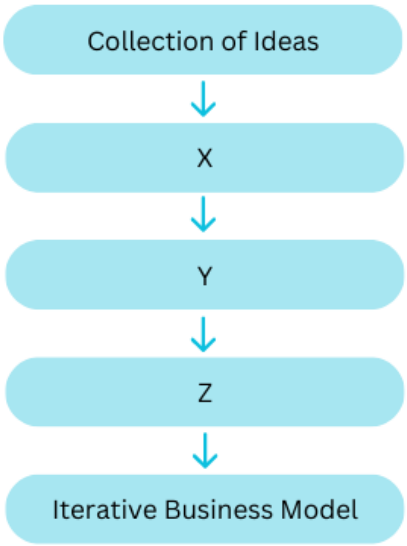
1. The challenges of innovation are at two levels- individual and organisational. Pick out the challenge faced at the organizational level. (select all that apply)

A. Fear of Failure
B. Lack of internal heroes
C. Lack of role clarity
D. Lack of recognition for efforts
E. No innovation strategy

**Answers: B, D, E**

Organizational-level challenges are lack of internal heroes, lack of recognition for efforts and no innovation strategy. These sit with the company, not the person.

2. Identify the missing X, Y, and Z steps in innovation management.




A. Assessment of Ideas, Selection of Ideas, Prototyping
B. Brainstorming, Designing, Validation
C. Selection of Idea, Experimentation with Idea, Developing the Business model

D. Conducting research, designing the identity, Developing the beta version

**Answer: C**

The flow is Collection of Ideas, then Selection of Idea (X), Experimentation with Idea (Y), Developing the Business Model (Z), then Iterative Business Model.

3. Select the correct statement regarding divergent thinking.

A. The process of thinking that explores multiple possible solutions in order to generate creative ideas is divergent thinking.
B. Divergent thinking helps in finding solutions to a problem through deducing by applying established rules and logical reasoning.
C. It generally means the ability to give the "correct" answer to standard questions that do not require significant creativity.
D. The term divergent thinking is coined by Joy Paul Guilford.

**Answer: A**

Divergent thinking explores many possible solutions to generate creative ideas.

4. Select the correct statement/s.
Statement 1: The objective of the Idea Management System is to overcome resistance to change.
Statement 2: The objective of the Idea Management System is to overcome the prediction disability.

A. Both statements are false.
B. Statement 1 is True, and Statement 2 is false.
C. Both statements are true.
D. Statement 1 is False, and Statement 2 is True.

**Answer: C**

Both are true. An Idea Management System is meant to overcome resistance to change and the prediction disability (not being able to predict which idea will succeed).

5. Select the key processes in the innovation programme. (select all that apply)

A. Idea Management
B. Buzz Creation
C. Implementing employee recreational activities
D. Vendor Management
E. Training and Development

**Answers: A, B, E**

Key processes in an innovation programme are Idea Management, Buzz Creation, and Training and Development.

6. Who said, "if there are challenges and those challenges are difficult, then some interesting, innovative solutions will come."

A. Elon Musk
B. Ratan Tata
C. Thomas Lockwood
D. Peter F Drucker

**Answer: B**

It was Ratan Tata who said difficult challenges bring out innovative solutions.

7. The concept of the challenge book helps in developing the Idea Bank.

A. The above statement is false.
B. The above statement is partially false.
C. The above statement is true.
D. The above statement is partially true.

**Answer: C**

The statement is true. The challenge book collects challenges that feed the Idea Bank.

8. Which source of challenge book is it when customers want high-quality and low-cost products?

A. Feel the pain source.
B. Sense the wave source.

C. See the waste source.
D. Perceive the change source.

**Answer: A**

Customers wanting high quality at low cost is a customer pain. That is the 'Feel the pain' source.

9. What are the three important dimensions (3Ps) of sustainability? (select all that apply)

A. Product
B. People
C. Participation
D. Planet
E. Profit
F. Process

**Answers: B, D, E**

The 3Ps of sustainability are People, Planet and Profit.

10. Natural resources, energy, technological resources, etc., are being wasted mindlessly. There is not enough power generation in India to meet the demands. An innovator is coming up with a solution to curb the problem of power generation. It is an example of which source of a challenge book:

A. Feel the pain source
B. Sense the wave source
C. See the waste source
D. None of the above

**Answer: C**

Wasted natural resources and energy is the 'See the waste' source.

## Week 3

1. An innovator wants to create an app which can help teenagers and young kids to safely use social media. What could be the possible assumptions to be validated in this scenario?
Types of Assumptions-
(a) Need
(b) Technical
(c) Production
(d) Commercial
Assumptions to be validated-
(i) There are sufficient number of parents who are ready to subscribe and pay for such an app.
(ii) There is a demand of an app to monitor and facilitate safe use of social media by teenagers and young kids.
(iii) There is expertise on child psychology and social media operations.
(iv) To have app version for different platforms like android, iOS etc.
Please choose the correct options below:

A. a-ii, b-i, c-iii, d-iv
B. a-i, b-iv, c-ii, d-iii
C. a-ii, b-iii, c-iv, d-i
D. a-i, b-iii, c-iv, d-ii

**Answer: C**

Match each assumption type to its example. Need = demand for the app (ii). Technical = expertise on child psychology and social media operations (iii). Production = app versions for Android, iOS etc. (iv). Commercial = parents ready to pay (i). That is a-ii, b-iii, c-iv, d-i.

2. Statement I: A key component of experimentation in innovation is to get the desired results every time.
Statement II: For successful validation of all the assumptions of an idea, designing a low-cost experiment is very crucial.
Which of the following is correct?

A. Only Statement I is correct.
B. Only Statement II is correct.
C. Both the statements are correct.

D. Both the statements are incorrect.

**Answer: B**

Statement II is true: a low-cost experiment is key to validating assumptions. Statement I is false: experiments are for learning, not for getting the desired result every time.

3. The main hinderance in the path of experimentation to validate an innovative idea is:

A. Low-cost experimental setup
B. To keep trying until desired results are achieved
C. The failure fallacy
D. All of the above

**Answer: C**

The main hindrance is the failure fallacy: treating a failed experiment as failure of the idea or the person instead of as learning.

4. Role/roles of an idea champion includes:

A. Building support for the new idea or technology
B. Help overcoming resistance
C. Make our idea sticky
D. All of the above

**Answer: D**

An idea champion does all of these: builds support for the idea, helps overcome resistance and helps make the idea sticky.

5. Which of the following is not a requirement to make an idea sticky?

A. Concrete
B. Concern
C. Curiosity
D. Credibility

**Answer: B**

Concern is not one of the requirements for a sticky idea.

6. Consider a situation faced by a team in an organization to come up with a solution to counter high competition and dropping sales of their product. One of the team members, Ajay comes up with an idea of radicle changes in the product design but is unable to communicate the idea convincingly. Another member Sunil not only believed in Ajay's idea but also tried to help overcome other member's resistance. Finally, the team leader, Jaya having a strong hold in the organization pushed forward the idea and helped implementing the idea by making people accept and work for the idea. Who do you think should be considered an idea champion in this case?

A. Ajay
B. Sunil
C. Jaya
D. All of the above

**Answer: C**

Jaya is the idea champion. She had the position to push the idea forward and got people to accept and implement it.

7. Having a "coding club" in a college would be an example of:

A. Learning environment
B. A team activity
C. Community of practice
D. A group of role models

**Answer: C**

A coding club is a group with a shared interest that learns from each other. That is a community of practice.

8. Which of the following could be ways of recognizing the efforts of the employees and encourage them to be innovative?

A. Giving cash incentives to bring in new ideas

B. Putting up employee's photo as "Innovator of the Month"
C. Taking most idea givers on a holiday retreat
D. All of the above

**Answer: D**

All three are ways to recognize and encourage innovative employees: cash incentives, public recognition and rewards like a retreat.

9. Statement I: Japanese organizations differentiate between value added activities and non-value added to become more efficient and create more value.
Statement II: Value creation is at the core of any economic exchange.
Which of the following is correct?

A. Only Statement I is correct.
B. Only Statement II is correct.
C. Both the statements are correct.
D. Both the statements are incorrect.

**Answer: C**

Both are true. Japanese firms separate value-added from non-value-added activity, and value creation is the core of any economic exchange.

10. Which of the following statements is correct regarding co-creation?

A. Co-creation is a linear process.
B. Lego creating a platform "Lego Ideas" inviting ideas and suggestions from over 1 million people where fans voted for the most popular ideas, would be an example of co-creation.
C. In co-creation process the burden of value addition lies with the producer.
D. Concept of co-creation is only applicable in service industry.

**Answer: B**

Lego Ideas is a real example of co-creation: customers and fans take part in creating value.

## Week 4

1. Select the correct statement/s.
Statement 1: Business scenarios may never be directed by business models.
Statement 2: The business scenario is directed by needs and integration or disintegration of people.

A. Both statements are false.
B. Statement 1 is True, and Statement 2 is false.
C. Both statements are true.
D. Statement 1 is False, and Statement 2 is True.

**Answer: D**

Statement 2 is true: business scenarios are directed by needs and by people coming together or splitting up.
Statement 1 is false: scenarios are directed by business models.

2. Which of the following is NOT a quality deemed necessary in an entrepreneur?

A. Analytical abilities and originality
B. Aptitude for Human Relations
C. Communication Skills
D. Physical appeal

**Answer: D**

Physical appeal is not a quality needed in an entrepreneur.

3. If a person organizes a business, manages a business, and takes risks, he best fits to be a/an:

A. Actor
B. Doctor
C. Entrepreneur
D. Teacher

**Answer: C**

A person who organizes a business, manages it and takes the risk is an entrepreneur.

4. Which among the following are the best examples of social entrepreneurship? (select all that apply)

A. Grameen Bank of Bangladesh
B. ITC Limited
C. Aravind Eye Hospital & Aurolab
D. Patanjali Ayurved
E. Study Hall Education Foundation

**Answers: A, C, E**

Grameen Bank, Aravind Eye Hospital with Aurolab, and Study Hall Education Foundation are built to solve social problems.

5. Select the correct statement/s regarding social entrepreneurship.
Statement 1: A social entrepreneur is a person who explores business opportunities that have a positive impact on their community, society, or the world.
Statement 2: Social entrepreneurship involves a person or group who owns their own for-profit business.
Statement 3: The process of developing and funding solutions that directly address social issues is known as social entrepreneurship.

A. All the statements are True.
B. Statement 1 and Statement 3 are true, and Statement 2 is false.
C. Statement 1 is False, and Statement 2 and Statement 3 are True.
D. Statement 1 and Statement 2 are true, and Statement 3 is false.

**Answer: B**

Statements 1 and 3 are true: social entrepreneurs seek opportunities that benefit society, and the work is about funding solutions to social issues. Statement 2 is false: owning a for-profit business does not define social entrepreneurship.

6. Daniel Kahneman is a well-known psychologist and Nobel Prize winner in Economics who authored the bestseller:

A. Start, Grow, and Run
B. The Innovator's Dilemma
C. Thinking, Fast and Slow
D. Zero to One

**Answer: C**

Kahneman wrote 'Thinking, Fast and Slow'.

7. Which of the following is not an example of social entrepreneurship?

A. Producing environmentally friendly products
B. Mass production of plastic bottles to turn a profit.
C. Providing banking services in underserved areas
D. Helping children orphaned by epidemic disease.

**Answer: B**

Mass-producing plastic bottles to make a profit has no social aim, so it is not social entrepreneurship.

8. Which among the following is NOT a step in six path framework involved in formulating the blue ocean strategy?

A. Look across alternative industries.
B. Look across strategic groups within industry.
C. Look across the chain of buyers.
D. Look across space.

**Answer: D**

The six paths are alternative industries, strategic groups, chain of buyers, complementary products and services, functional-emotional orientation, and time. 'Look across space' is not a path.

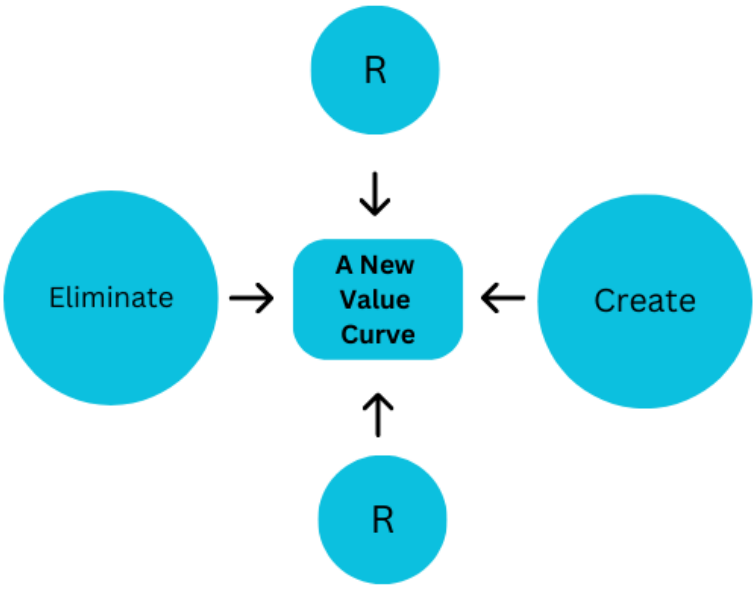
9. What are the four steps of visualizing strategy?

A. Visual Awakening, Visual Exploration, Visual Strategy Fair, Visual Communication
B. Stimulus, Sensor, Efferent, Response
C. Objectives, Formulation, Implementation, Evaluation and Control
D. Product, Place, Promotion, Price

**Answer: A**

The four steps of visualizing strategy are Visual Awakening, Visual Exploration, Visual Strategy Fair and Visual Communication.

10. Identify the missing elements in the Four action framework of the blue ocean strategy.




A. Reduce and Reuse
B. Reduce and Raise
C. Recycle and Recreate
D. Recycle and Raise

**Answer: B**

The Four Actions Framework is Eliminate, Reduce, Raise, Create. The missing elements are Reduce and Raise.

## Week 5

1. Which of the following is correct?

A. Engineering->Technology->Science
B. Science->Engineering->Technology
C. Science->Technology->Engineering
D. Technology->Engineering->Science

**Answer: B**

The order is Science, then Engineering, then Technology.

2. Statement I: All scientific inventions lead to technological innovation.
Statement II: Technological innovation is when an invention is transformed into a socially usable product.
Which of the following is correct?

A. Only Statement I is correct.
B. Only Statement II is correct.
C. Both the statements are correct.
D. Both the statements are incorrect.

**Answer: B**

Statement II is true: technological innovation is an invention made into a socially usable product. Statement I is false: not every scientific invention becomes a technological innovation.

3. Which of the following is not true regarding technological innovation?

A. It is about creating new use of the existing technology.
B. It is a complex process involving technological, social and economic factors.
C. It requires huge financial investments.
D. It takes long period of time, sometimes decades for such innovations to be impactful.

**Answer: A**

The question asks for the NOT true statement. Technological innovation is not just finding a new use for existing technology. It means turning new science and engineering into technology that society can use.

4. Which of the following are associated with technological innovation process:
(a) Basic research
(b) Product Development
(c) Societal impact- positive and negative
(d) Co-creation
(e) Excluding existing knowledge
(f) Scaling up
(g) Trial and error
(h) Experimentation
Please select the correct option:

A. (a), (b), (d), (f), (h)
B. (a), (b), (c), (d), (f)
C. (a), (b), (e), (f), (g)
D. (b), (d), (e), (f), (h)

**Answer: A**

Associated with the technological innovation process: basic research (a), product development (b), co-creation (d), scaling up (f) and experimentation (h).

5. Technological innovation can be divided into following catagories:
Types-
(a) Radical innovation
(b) Micro-radical innovation
(c) Generational innovation
(d) Incremental innovation
Examples-
(i) Virtual trial rooms
(ii)Smartphone camera enhancement
(iii) Creation of world wide web
(iv) Transition from 3G to 4G mobile network
Please select the correct match below:

A. a-ii, b-i, c-iii, d-iv
B. a-i, b-iv, c-ii, d-iii
C. a-iii, b-i, c-iv, d-ii
D. a-iv, b-iii, c-i, d-ii

**Answer: C**

Radical = creation of the world wide web (iii). Micro-radical = virtual trial rooms (i). Generational = 3G to 4G transition (iv). Incremental = smartphone camera enhancement (ii). That is a-iii, b-i, c-iv, d-ii.

6. Which of the following is not true:

A. S-curve helps stakeholders anticipate the trajectory of a technology's development and plan for future developments and challenges.
B. Incremental innovation involves short term planning while micro-radicle and generational innovation involves long term planning.
C. Radical innovations are rare phenomenon and are difficult to plan based on S-curve.
D. India majorly has 3rd generation companies where R&D; planning is embedded in corporate and business planning.

**Answer: D**

The question asks for the NOT true statement. India mostly does not have third-generation companies with R&D; built into corporate planning. So this statement is the untrue one.

7. Based on competitive impact, technologies can be divided as:
(a) Base technologies
(b) Key technologies
(c) Pacing technologies
(d) Emerging technologies
Match with the correct statements-
(i) Technologies well embodied in products and are basis of competitive advantage as they provide differentiation to companies.
(ii) Technologies which already demonstrated their ability to change the basis of competition and as a result are now moved to the development stage.
(iii) Ageing technology which is available or known to all industry participants but cannot provide competitive advantage because of its widespread use.
(iv) Forward looking technologies at an early research stage with unknown potential, but promising and could provide competitive advantage in future.
Please select the correct option below:

A. a-i, b-ii, c-iii, d-iv
B. a-iii, b-i, c-ii, d-iv
C. a-iii, b-i, c-iv, d-ii
D. a-i, b-iii, c-ii, d-iv

**Answer: B**

Base = ageing technology known to all (iii). Key = embodied in products, basis of differentiation (i). Pacing = shown ability to change competition, now moving to development (ii). Emerging = early research with unknown potential (iv). That is a-iii, b-i, c-ii, d-iv.

8. Which of the following statements are true for strategy?

A. Offensive strategies help maintain the status quo and usually help expand the existing markets.
B. Absorbent strategies involve acquiring license to exploit innovation in domestic market.
C. Dependent strategies focus on developing core strategy with the parent company.
D. Imitative strategies are research intensive strategies.

**Answer: B**

Absorbent strategies involve acquiring a licence to exploit an innovation in the domestic market.

9. Which of the following are technology forecasting tools?
(a) Time series analysis
(b) Morphological analysis
(c) Delphi technique
(d) Moving average
(e) Pearl function based on S-curve
(f) Envelop curve and trend extrapolation
Please select the correct option:

A. (a), (d), (e), (f)
B. (a), (b), (c), (e)
C. (b), (c), (e), (f)
D. (b), (d), (e), (f)

**Answer: C**

The forecasting tools in the course key are morphological analysis (b), the Delphi technique (c), the Pearl function based on the S-curve (e), and envelope curve with trend extrapolation (f).

10. Statement I: Technology forecasting is important for businesses to anticipate any future technological developments, trends, innovations and to develop strategies to take advantage of future opportunities and mitigate any potential threat to the existing business.
Statement II: These forecasting tools assist organizations in making informed decisions about research, development, investment, and strategic planning.
Which of the following is correct?

A. Only Statement I is correct.
B. Only Statement II is correct.
C. Both the statements are correct.
D. Both the statements are incorrect.

**Answer: C**

Both are true. Forecasting helps firms anticipate trends and plan strategy, and the tools support decisions on research, development and investment.

## Week 6

1. The definition of Innovation given by Robenstein states that “Innovation is the process whereby new and improved _____ are developed and transferred to a new plant or market where they are appropriate.”

A. Products, processes, materials, and services
B. Goods and services
C. Idea, method, or device
D. Designs and business models

**Answer: A**

Robenstein defines innovation as new and improved products, processes, materials and services being developed and transferred to a new plant or market.

2. Who gave the concept of the 5 Cs of insight development?

A. Ratan Tata
B. Robenstein
C. Thomas Lockwood
D. Gary Klein

**Answer: D**

Gary Klein gave the 5 Cs of insight development.

3. Which of the following example illustrates sustainable Innovation the best?
Example 1: Arvind eye care hospital
Example 2: Narayana Hrudayalaya (Narayana Health), an Indian chain of multi-specialty hospitals, heart centers, and primary care facilities.

A. Example 1
B. Example 2
C. Both a and b
D. None of the above

**Answer: C**

Both examples show sustainable innovation: they serve society at large scale and keep running financially. So 'both' is right.

4. Which of the following is an example of societal development with reference to sustainability?

A. Arvind eye care hospital
B. Narayana Hrudayalaya
C. River Narmada cleanliness drive named “Narmada ji ka Safai Abhiyan”
D. A grass-cutting machine uses coal for energy.

**Answer: C**

The Narmada cleanliness drive is a community-driven social initiative. That is societal development.

5. Example of bar codes being used for inventory control will fall under which Product/Process – Usage/Problem matrix category?

A. Category 1 (New Product/Process – Old Usage/Problem)
B. Category 2 (Old Product/Process – Old Usage/Problem)
C. Category 3 (New Product/Process – New Usage/Problem)
D. Category 4 (Old Product/Process – New Usage/Problem)

**Answer: C**

Bar codes for inventory control are a new product or process used for a new usage or problem. That is Category 3.

6. The limits of investment in plant and machinery for a medium organization is ____.

A. Rs 10 lakh rupees 1 million
B. Rs 25 lakh or 2.5 million
C. Rs 5 crore or 50 million
D. Rs 10 crore or 100 million

**Answer: D**

The course key gives Rs 10 crore (Rs 100 million) as the investment limit in plant and machinery for a medium enterprise.

7. What do you mean by ASPIRE?

A. A Scheme for Promoting Innovation and Rural Entrepreneurship
B. A systematic plan to Implement Rural Entrepreneurship
C. A Single Point of Institutional Resource Entry
D. Asian School of Planning Innovation and Rural Entrepreneurship

**Answer: A**

ASPIRE stands for A Scheme for Promoting Innovation and Rural Entrepreneurship.

8. KVIC stands for:

A. Khadi and Village Industries Committee
B. Kerala Village Innovation Council
C. Khadi Village Innovation Council
D. Knowledge Value Innovation Culture

**Answer: A**

The course key marks 'Khadi and Village Industries Committee' for KVIC. (In real life the body is the Khadi and Village Industries Commission, but this is the option the key accepts.)

9. Which among these is an example of Innovation resulting from social initiatives?

A. Girl child friendly school infrastructure
B. Use of sustainable materials
C. Reduced expenses
D. Ability to meet growing demand

**Answer: A**

Girl-child-friendly school infrastructure is an innovation from a social initiative.

10. Which among the following is the benefit of innovations resulting from environmental initiatives?

A. The exploitation of local biodiversity
B. Reduction of environmental impact at the source ( e.g., recovery of the paper waste)
C. Use of non-sustainable materials such as single-use plastic
D. None of the above

**Answer: B**

Reducing environmental impact at the source, such as recovering paper waste, is a benefit of environmental initiatives.

## Week 7

1. _____ is the sum total of all the knowledge in an enterprise in the minds of it's employees which can be used to create wealth.

A. Intellectual Capital
B. Intellectual Asset
C. Intellectual Property
D. Knowledge-based resources

**Answer: A**

Intellectual capital is the sum of all the knowledge in the minds of employees that can be used to create wealth.

2. Which of the following is correct regarding creation of intellectual capital?

A. Purpose->Information->Knowledge->Strategy
B. Purpose->Strategy->Information->Knowledge
C. Information->Knowledge->Purpose->Strategy
D. Information->Purpose->Strategy->Knowledge

**Answer: B**

The order is Purpose, then Strategy, then Information, then Knowledge.

3. Statement I: Intellectual assets are intellectual capital which is identified, documented and available to be shared and replicated within the organization.
Statement II: Imagine a software company that specializes in developing customer relationship management (CRM) software for businesses. They have created a cutting-edge CRM application that offers advanced features, data analytics, and user-friendly interfaces. The relationships developed with the customers is a valuable intellectual asset for the company.
Which of the following is correct?

A. Only Statement I is correct.
B. Only Statement II is correct.
C. Both the statements are correct.
D. Both the statements are incorrect.

**Answer: A**

Statement I is true: intellectual assets are identified, documented capital that can be shared and replicated. Statement II is false: customer relationships live with people. They are intellectual capital, not a documented, replicable intellectual asset.

4. Intellectual Capital->Intellectual Assets-> _____

A. Patents
B. Knowledge-based Economy
C. Intellectual Property
D. Intellectual Ownership

**Answer: C**

Intellectual capital becomes intellectual assets, which get legal protection as intellectual property.

5. Statement I: Intellectual property rights and patents can be interchangeably used.
Statement II: Legal protection of new creations leads to monopoly and discourages commitment of additional resources for further innovation.
Which of the following is correct?

A. Only Statement I is correct.
B. Only Statement II is correct.
C. Both the statements are correct.
D. Both the statements are incorrect.

**Answer: D**

Both are false. IPR is a wider set (copyright, patents, trademarks and more), so the two are not interchangeable. Legal protection rewards creators and encourages further innovation, rather than discouraging it.

6. According to WIPO, the IPR system helps strike a balance between interests of _____ and _____ providing an environment in which creativity and invention can flourish, for the benefit of all.

A. Industry and public
B. Innovators and industry
C. Innovators and public
D. Industry and economy

**Answer: C**

According to WIPO, the IPR system balances the interests of innovators and the public.

7. Match the following: Type of IPR-
(a) Copyright

(b) Patent
(c) Trademark
Examples-
(i) A novelist's original manuscript
(ii) The "Nike" logo and brand name
(iii) Pharmaceutical company obtaining the exclusive right to produce and sell the medication for a specified period.
Please select the correct option below:

A. a-i, b-ii, c-iii
B. a-i b-iii, c-ii
C. a-ii, b-i, c-iii
D. a-iii, b-ii, c-i

**Answer: B**

Copyright = a novelist's manuscript (i). Patent = exclusive right to produce and sell a medication for a period (iii). Trademark = the Nike logo and brand name (ii). That is a-i, b-iii, c-ii.

8. Which of the following falls into the IPR category?

A. The distinctive design of a luxury car's exterior, including its shape, curves, and exterior features
B. The secret recipe for a famous soft drink
C. Darjeeling Tea is a tea that can only be labeled as such if it originates from Darjeeling region
D. All of the above

**Answer: D**

All three fall under IPR: a distinctive design (industrial design), a secret recipe (trade secret) and Darjeeling Tea (geographical indication).

9. A scientist develops a new biodegradable packaging material made from a blend of agricultural waste fibers and biopolymers. Prior to the development of the biodegradable packaging material, experts in the field were using traditional plastics for packaging. The scientist's innovative combination of agricultural waste fibers and biopolymers represents a departure from conventional practices. The biodegradable packaging material developed by the scientist serves the practical utility of providing an environmentally friendly alternative to traditional plastic packaging. To demonstrate the usefulness of the innovation, the scientist conducted a seminar. Which of the following is true regarding this scenario?

A. The scientist will be given a patent for a limited period.
B. The scientist’s patent will be limited to the region she is residing in.
C. The scientist will not be getting any protection on her innovation.
D. None of the above.

**Answer: C**

The scientist held a seminar to show the invention, which is a public disclosure. After that the idea is no longer new, so no patent protection is available.

10. Statement I: Copyright is a legal protection granted to the creators of original literary, artistic, and creative works. It gives the creators exclusive rights to control how their works are used, reproduced, distributed, performed, or displayed.
Statement II: Copyright is the protection of the original idea so that expression of that idea can be controlled.
Which of the following is correct?

A. Only Statement I is correct.
B. Only Statement II is correct.
C. Both the statements are correct.
D. Both the statements are incorrect.

**Answer: A**

Statement I is true: copyright gives creators exclusive rights over use of their original works. Statement II is false: copyright protects the expression of an idea, not the idea itself.

## Week 8

1. _____ is a strategic framework that outlines how a company creates, delivers, and captures value in the market.

A. Value proposition
B. Business Model
C. Business Strategy
D. Revenue Model

**Answer: B**

A business model is the framework that explains how a company creates, delivers and captures value.

2. An online fitness platform aims to provide a convenient and accessible way for individuals to engage in fitness activities and achieve their health goals. The business model of the platform includes the following components:
(a) Target customers
(b) Value proposition
(c) Competition
(d) Revenue streams and costs
(e) Key Activities
(f) Promotion and distribution channels
(g) Government regulations
(h) Review and feedback
Please select the correct option:

A. (a), (b), (d), (g), (h)
B. (a), (b), (c), (e), (f)
C. (a), (b), (d), (e), (f)
D. (b), (d), (e), (f), (h)

**Answer: C**

The business model components are target customers (a), value proposition (b), revenue streams and costs (d), key activities (e), and promotion and distribution channels (f).

3. Statement I: One of the major reasons for business models to fail is when one or more elements or components of the model aren't working as they are supposed to.
Statement II: Businesses inability to adapt in response to dynamic changing environment and competitive landscape also leads to business failure.
Which of the following is correct?

A. Only Statement I is correct.
B. Only Statement II is correct.
C. Both the statements are correct.
D. Both the statements are incorrect.

**Answer: C**

Both are true. Business models fail when elements stop working as intended, and also when the firm cannot adapt to a changing environment.

4. Which of the following would be considered as remedies to address the failures a business is facing?

A. Having a futuristic approach where the business try to anticipate the features or characteristics of their product or services or processes would be desirable by the consumers.
B. Focus on the areas of consideration to understand the symptoms of the failure and thereby trying to eliminate its causes by choosing alternate courses of action.
C. Matching the capacity and capabilities with the value proposition.
D. All of the above

**Answer: D**

All three are remedies: anticipating what consumers will want, finding the causes behind the symptoms of failure, and matching capacity and capabilities with the value proposition.

5. Which of the following statement is not correct?

A. Entrepreneurship is a chain of events creating, developing, and managing a business venture with the aim of identifying opportunities, taking calculated risks, and generating value.

B. Business incubators are supportive environments designed to nurture the growth and development of early-stage startups and entrepreneurs.
C. In practice, incubators provide handholding throughout the entire entrepreneurial journey from pre-founding stage to early development stage.
D. Incubators offer a range of resources, mentorship, and infrastructure to help startups overcome challenges, refine their business models, and accelerate their journey.

**Answer: C**

The question asks for the statement that is NOT correct. Per the course key, C is incorrect: incubator support is time-limited and early-stage, not handholding through the entire entrepreneurial journey.

6. Which of the following is NOT a common feature of business incubators?

A. Networking opportunities
B. Sole focus on established companies
C. Access to shared resources
D. Mentorship and expert guidance

**Answer: B**

Incubators help early-stage startups. A sole focus on established companies is not a feature of them.

7. What is the usual time frame for startups to stay within a business incubator program?

A. A few decades
B. A year or two
C. A few weeks
D. Indefinitely

**Answer: B**

Startups usually stay in an incubator for a year or two.

8. What is the primary source of funding for most startups in their early stages?

A. Government grants
B. Angel investors
C. IPOs
D. Self

**Answer: B**

The official course key marks Angel investors as the main early-stage funding source. (The slide lists self-funding first, but the key is what the exam follows.)

9. Statement I: Debt financing is now also possible against intellectual property as collateral.
Statement II: Debt financing the first level of financing for startups, financing through equity follows in the later stages.
Which of the following is correct?

A. Only Statement I is correct.
B. Only Statement II is correct.
C. Both the statements are correct.
D. Both the statements are incorrect.

**Answer: A**

Statement I is true: debt financing against intellectual property as collateral is now possible. Statement II is false: for startups equity usually comes first and debt comes in later stages.

10. Which of the following are some key areas which require innovation and help create future markets?
(a) Technology and digital transformation
(b) Renewable energy and sustainability
(c) Healthcare and biotechnology
(d) Disaster management
(e) Ed-tech
(f) Fintech and financial inclusion
(g) Agriculture and food security
(h) Urbanization and smart cities
(i) Defense

Please select the correct option:

A. (b), (c), (g)
B. (a), (d), (e), (f), (i)
C. Only (b) and (g)
D. (a), (b), (c), (d), (e), (f), (g), (h), (i)

**Answer: D**

All nine areas need innovation and help create future markets, from digital technology to defense.

## Debatable questions

Answers above follow the NPTEL portal accepted answers everywhere. The questions below are the ones where the official key is debatable or the official solution PDF gives no key. For the exam, always answer the portal version.

**W2 Q3** - Select the correct statement regarding divergent thinking.

**Portal accepted answer:** A - the process of thinking that explores multiple possible solutions to generate creative ideas.

**Official solution PDF:** A (bold-marked in official Week 2 solution PDF) - same as portal.

**Lecture-based view:** Agree with A - it matches the lecture definition. Caveat: option D (the term was coined by Joy Paul Guilford) is also factually true, so the question design is loose. Exam answer: A.

**W5 Q3** - Which of the following is not true regarding technological innovation?

**Portal accepted answer:** A - it is about creating new use of the existing technology.

**Official solution PDF:** No marked answers in the official Week 5 PDF (question paper only).

**Lecture-based view:** Agree with A: the week 5 lecture says technological innovation is complex, takes decades, and needs substantial investment - so B, C, D are all true, leaving A as the not-true one. Mildly debatable because 'new use of existing technology' is a common plain-language description of innovation.

**W5 Q4** - Which of the following are associated with the technological innovation process?

**Portal accepted answer:** (a), (b), (d), (f), (h) - basic research, product development, co-creation, scaling up, experimentation.

**Official solution PDF:** No marked answers in the official Week 5 PDF.

**Lecture-based view:** Agree, but debatable: societal impact (c) is also discussed in the lecture as part of adoption, so option B (which swaps experimentation for societal impact) is defensible. Portal accepts A.

**W6 Q5** - Bar codes used for inventory control fall under which Product/Process - Usage/Problem matrix category?

**Portal accepted answer:** Category 3 (New Product/Process - New Usage/Problem).

**Official solution PDF:** Category 3 (official Week 6 answer table: Q5 = C) - same as portal.

**Lecture-based view:** Arguably Category 4: bar codes are an old technology applied to a (then) new usage. The lecture itself admits the line between these categories is thin. Exam answer: Category 3.

**W7 Q9** - Scientist publicly presented her biodegradable packaging material at a seminar before filing - patent protection?

**Portal accepted answer:** C - the scientist will not be getting any protection on her innovation.

**Official solution PDF:** No marked answers in the official Week 7 PDF.

**Lecture-based view:** Mostly agree: public disclosure before filing destroys novelty. Not absolute - Indian law has narrow grace exceptions (recognised exhibitions, papers read before a learned society), so a vague 'seminar' could go either way. Exam answer: C.

**W8 Q5** - Which statement is not correct? (about incubators)

**Portal accepted answer:** C - in practice, incubators provide handholding throughout the entire entrepreneurial journey from pre-founding to early development stage.

**Official solution PDF:** No marked answers in the official Week 8 PDF.

**Lecture-based view:** Agree C is the not-correct statement: incubation is time-limited (startups graduate in a year or two, per the same week's lecture), so 'entire entrepreneurial journey' overshoots. Slightly ambiguous wording since pre-founding to early development is genuinely the incubator span.

**W8 Q8** - What is the primary source of funding for most startups in their early stages?

**Portal accepted answer:** B - Angel investors.

**Official solution PDF:** No marked answers in the official Week 8 PDF.

**Lecture-based view:** Disagree-ish: the week-8 slide lists 'Self' first under sources of investment, and in practice most early-stage startups are self-funded (founders, friends and family). Portal wants Angel investors - for the exam, answer Angel investors.